



International
Grammar School

2008
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

Extension 2

General Instructions

- Reading Time- 5 minutes
- Working Time – 3 hours
- Write using a blue or black pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.

Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value

Total Marks – 120

Attempt Questions 1-8

All Questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE answer booklet

Marks

a) Find $\int \frac{dx}{\sqrt{16-9x^2}}$ **2**

b) Find $\int 5\cos x \sin^2 x \, dx$ **2**

c) Evaluate $\int_1^e x \ln x \, dx$ **3**

d) Evaluate $\int_2^3 \frac{dx}{x^2-1}$ **4**

e) Using the substitution $t = \tan \frac{\theta}{2}$ or otherwise find **4**

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$$

End of Question 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

a) Let $A = 3 + 4i$ and $B = 2 - 2i$. Find in the form $x + iy$ (x and y real).

i) $\frac{A}{B}$ 2

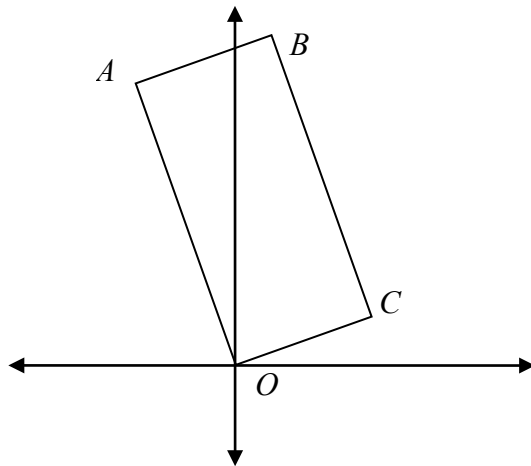
ii) $\sqrt{A}$ 3

iii) $A - \bar{B}$ 1

b) i) Write $1 + \sqrt{3}i$ in the form $r(\cos \theta + i \sin \theta)$ 2

ii) Hence write $(1 + \sqrt{3}i)^6$ showing that it is real. 2

c)



The points $OABC$ are the vertices of a rectangle on the Argand diagram with $|OA| = 2|OC|$. If OC represents the complex number $p + iq$, write down the complex numbers represented by:

i) OA 1

ii) OB 1

iii) BC 1

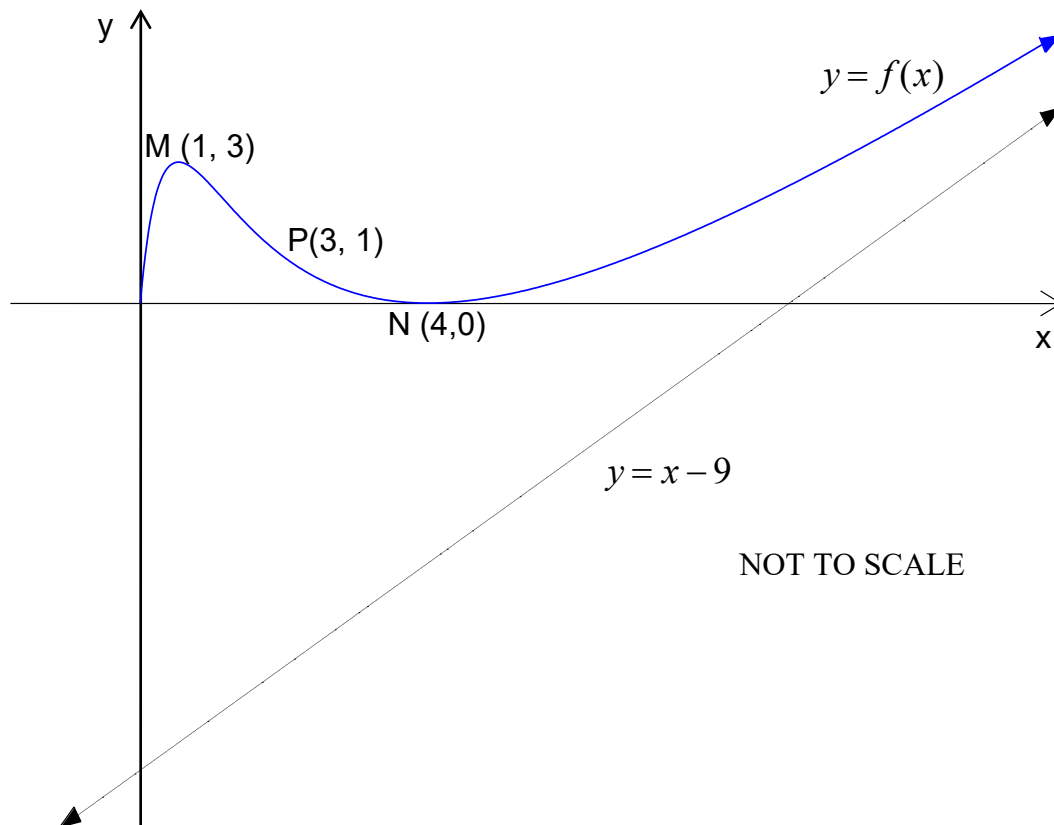
iv) AC 2

End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

a)



The diagram shows the graph of $y = f(x)$ for $x \geq 0$.

$M(1, 3)$ and $N(4, 0)$ are stationary points of $y = f(x)$ and $P(3, 1)$ is a point of inflexion of $y = f(x)$. The line $y = x - 9$ is an asymptote as $x \rightarrow \infty$. Draw separate one third page sketches showing any special features for the following:

i) $f'(x)$ 2

ii) $\frac{1}{f(x)}$ 2

iii) $-(f(x))^2$ 2

Question 3 continues on the next page

Question 3 continued**Marks**

- b) Determine the gradient of the tangent to the curve $x^2 + 2xy - y^2 = 17$ at the point $(3, 2)$ **2**
- c) The zeros of $x^3 - 3x^2 - 2x + 4$ are α , β and γ
- i) Find a cubic polynomial whose zeros are α^2 , β^2 and γ^2 **2**
- ii) Hence or otherwise find the value of $\alpha^2 + \beta^2 + \gamma^2$ **1**
- iii) Determine the value of $\alpha^3 + \beta^3 + \gamma^3$ **2**
- d) The equation $P(x) = x^3 + 3x^2 - 24x + k = 0$ has a double root. Find the possible values of k . **2**

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

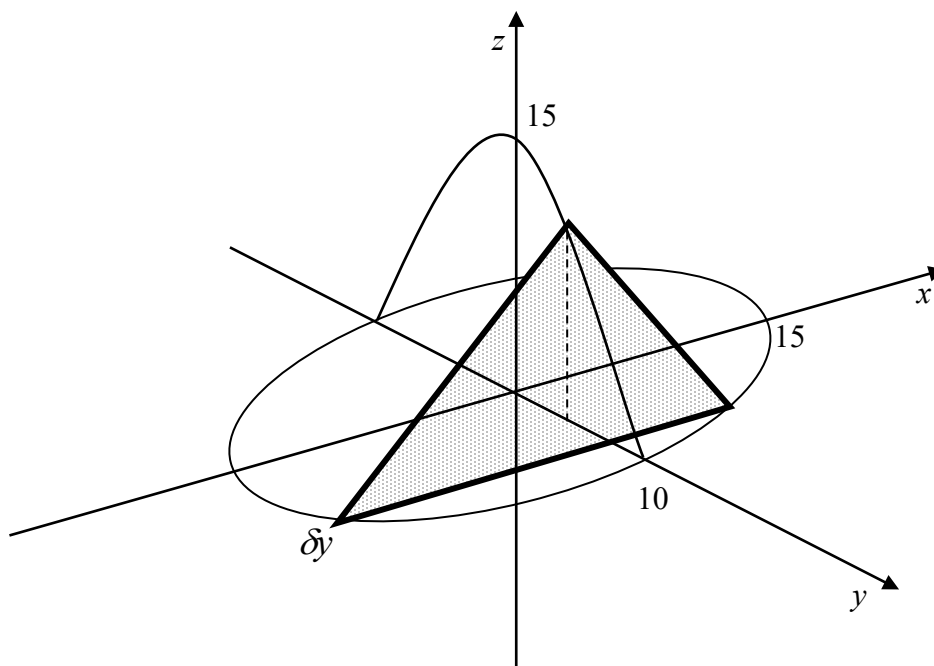
- a) i) Show that a reduction formula for $I_n = \int (\ln x)^n dx$ 3

$$\text{is } I_n = x(\ln x)^n - nI_{n-1}$$

- ii) Hence evaluate $\int_1^{e^4} (\ln x)^3 dx$ 4

- b) The arc of the curve $y = 6x - x^2 - 8$ where $y \geq 0$ is rotated about the line $x = 1$. 4
By applying the technique of cylindrical shells determine the exact volume of the solid formed

c)



A cake is made with base in the shape of an ellipse, with semi-major axis 15 cm and semi-minor axis 10 cm. Slices of the cake parallel to the major axis of the base are isosceles triangles, whose vertices trace out a semi-elliptical path with the same semi-major axis and semi-minor axis lengths as in the diagram below.

- i) Show that the volume of a ‘typical’ triangular slice is given by: 1
$$V_{\text{slice}} \approx xz\delta y \text{ cm}^3$$
- ii) Find the exact volume of the cake in cm^3 . 3

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) The line $y = mx + a$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at two points which have x coordinates x_1 and x_2 . **3**
- i) Express x_2 in terms of m, a, b and x_1 . **1**
- ii) Hence or otherwise show that the line is a tangent to the ellipse at the point where $\frac{-a^3m}{b^2 + a^2m^2}$. **1**
- b) A parabola has parametric equations $x = 2at$ and $y = at^2$. **2**
- i) Find the equation of the normal to the parabola at the point where $t = p$. **2**
- ii) Hence show that, through the point (x_1, y_1) , it is possible to draw up to three normals to the parabola. **2**
- c) Given the complex number $z = \cos \theta + i \sin \theta$ **3**
- i) Use De Moivre's Theorem and the binomial expansion find an expression for $\cos 4\theta$ in terms of $\cos \theta$ **2**
- ii) Also, using $z^n + \frac{1}{z^n} = 2 \cos n\theta$ determine an expansion for $\cos^4 \theta$ in terms of $\cos n\theta$ **2**
- iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 \theta \, d\theta$ **2**

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.**Marks**

- a) On a suitably labelled Argand diagram sketch the region determined by $[\operatorname{Re}(z)]^2 + \operatorname{Im}z < 0$

2

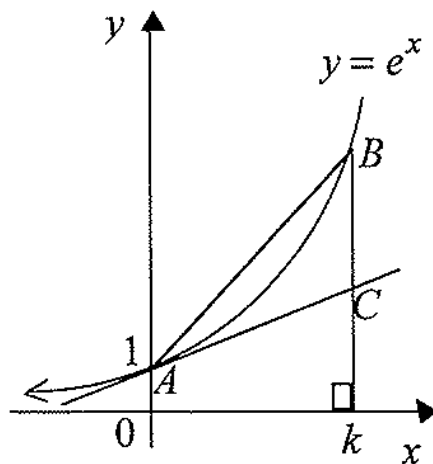
- (b) Consider the function

$$f(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

- (i) Use differentiation to show that $e^{-x} + x - 1 \geq 0$ for all values of x . Hence show that $f(x)$ is an increasing function for $x \neq 0$
- (ii) Show that $f(x)$ is continuous at $x = 0$.
- (iii) Sketch the graph of $y = f(x)$.

3**2****1**

- (c)



The curve $y = e^x$ cuts the y axis at A. B is a second point on the curve such that $x = k$, where $k > 0$. The tangent to the curve $y = e^x$ at A cuts the vertical line $x = k$ at the point C.

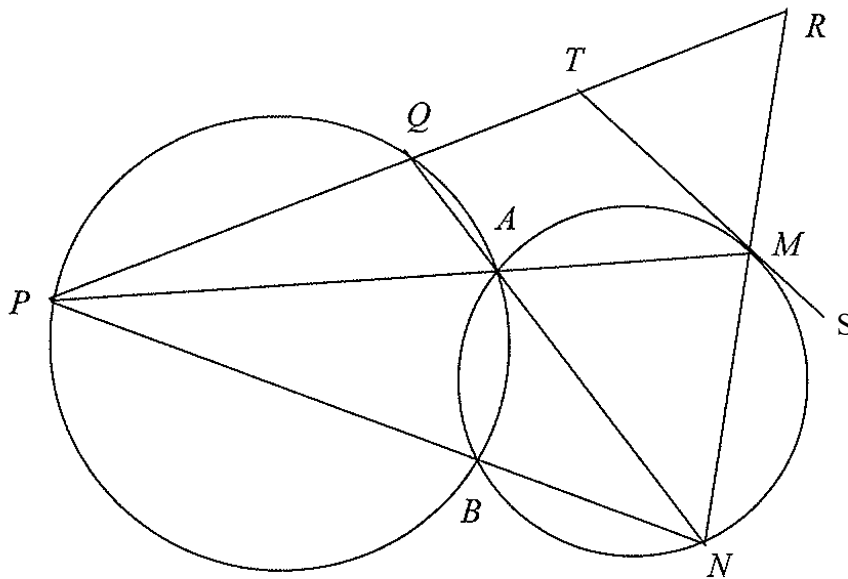
- (i) By considering areas, show that $\frac{1}{2}k(k+2) < e^k - 1 < \frac{1}{2}k(1+e^k)$. Hence deduce that $2.5 < e < 3$.
- (ii) Show that the curve $y = e^x$ bisects the area of $\triangle ABC$ for some value of k such that $2 < k < 3$. Taking $k = 2.7$ as a first approximation, apply Newton's method once to obtain a second approximation. Give your answer to one decimal place.

4**3****End of Question 6**

Question 7 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) Given that $z^5 - 1 = 0$
- Solve for Z over the complex field in the form $\cos \theta + i \sin \theta$. 3
 - Hence express $z^5 - 1$ as the product of linear and quadratic factors. 2
 - Write down the complex roots of $z^4 + z^3 + z^2 + z + 1 = 0$. 1
 - Without evaluating, show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ 2
- b)



In the diagram, the two circles intersect at A and B. P is a point on one circle. PA and PB produced meet the other circle at M and N respectively. NA produced meets the first circle at Q. PQ and NM produced meet at R. The tangent at M to the second circle meets PR at T. Copy or trace the diagram into your answer booklet.

- Show that QAMR is a cyclic quadrilateral. 3
- Show that $TM = TR$. 4

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet**Marks**

- a) i) Show that for all values of x and y : **1**
$$\sin(x+y) - \sin(x-y) = 2 \cos x \sin y$$
- ii) Use mathematical induction to show that for all positive integers n : **4**
$$\cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin\left(n + \frac{1}{2}\right)x - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$
- iii) Hence show that: **4**
$$\cos 2x + \cos 4x + \cos 6x + \dots + \cos 16x = 8 \cdot \cos 9x \cdot \cos 4x \cdot \cos 2x \cdot \cos x$$
- b) Show that a relationship between the coefficients of $p(x) = x^3 + ax^2 + bx + c = 0$ **4**
is $2a^3 - 9ab - 27c = 0$, if the roots are three consecutive terms of an arithmetic series.
- c) Solve the differential equation $\frac{dy}{dx} = 2y$ for y given that when $x = 1, y = 1$ **2**

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 Trial HSC Examination- Mathematics 2008
Extension 2

Part Solution

Marks Comment

- (a) $\int \frac{dx}{\sqrt{16-9x^2}} = \frac{1}{3} \int \frac{dx}{\sqrt{\frac{16}{9}-x^2}}$ 2
1 for rearranging
- $$= \frac{1}{3} \sin^{-1} \frac{x}{\frac{4}{3}} + c$$
- $$= \frac{1}{3} \sin^{-1} \frac{3x}{4} + c$$
- 1 for inv trig integral
- (b) $\int 5 \cos x \sin^2 x \, dx = \frac{5}{3} \sin^3 x + c$ 2
2 for solution
1 if simple error made
- (c) $\int_1^e x \ln x \, dx = \left[\frac{x^2}{2} \ln x \right]_1^e - \int_1^e \frac{x^2}{2} \frac{1}{x} dx$ 3
1 for breakup into parts
- $$= \left[\frac{x^2}{2} \ln x \right]_1^e - \int_1^e \frac{x}{2} dx$$
- $$= \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_1^e$$
- $$= \left[\frac{x^2}{4} (2 \ln x - 1) \right]_1^e$$
- $$= \frac{e^2}{4} (2-1) - \frac{1}{4} (-1)$$
- $$= \frac{e^2}{4} + \frac{1}{4}$$
- 1 for integral
- (d) Let $\frac{A}{x+1} + \frac{B}{x-1} = \frac{1}{x^2-1}$ 4
1 value of B
- $$A(x-1) + B(x+1) = 1$$
- When $x=1$ $2B=1 \rightarrow B=\frac{1}{2}$ 1 value of A
- When $x=-1$ $-2A=1 \rightarrow A=-\frac{1}{2}$
- $$\int_2^3 \frac{dx}{x^2-1} = \frac{1}{2} \int_2^3 \left(\frac{-1}{x+1} + \frac{1}{x-1} \right) dx$$
- $$= \frac{1}{2} [\ln(x-1) - \ln(x+1)]_2^3$$
- $$= \frac{1}{2} \left[\ln \left(\frac{x-1}{x+1} \right) \right]_2^3$$
- $$= \frac{1}{2} \left[\ln \frac{1}{2} - \ln \frac{1}{3} \right]_2^3$$
- $$= \frac{1}{2} \ln \frac{3}{2}$$
- 1 for answer

Question 1 Trial HSC Examination- Mathematics 2008
Extension 2

Marks Comment

Part Solution

4

(e) If $t = \tan \frac{\theta}{2}$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$$

$$2 \cos^2 \frac{\theta}{2} dt = d\theta$$

$$\cos^2 \frac{\theta}{2} = \frac{1}{1+t^2}$$

$$d\theta = \frac{2}{1+t^2} dt$$

Limits $\theta = \frac{\pi}{2} \rightarrow t = 1$

$\theta = 0 \rightarrow t = 0$

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta} = \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1+t^2}{(2+2t^2+1-t^2)} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{(3+t^2)} dt$$

$$= 2 \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}}$$

1 for $d\theta$

1 for correct
statement of
integral
including
limits

1 for
completing
integral

1 for result

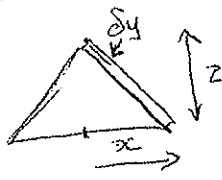
Question 2		Trial HSC Examination- Mathematics Extension 2	2008	
Part	Solution	Marks	Comment	
(a) i)	$\frac{a}{b} = \frac{3+4i}{2-2i} \times \frac{2+2i}{2+2i}$ $= \frac{6+6i+8i-8}{4+4}$ $= \frac{-2+14i}{8}$ $= -\frac{1}{4} + \frac{7}{4}i$	2	1 for multiplying by conjugate. 1 for correct answer	
ii)	Let $\sqrt{A} = x+iy$ $\therefore A = x^2 - y^2 + 2xyi$ $\therefore 3+4i = x^2 - y^2 + 2xyi$ $\therefore 3 = x^2 - y^2 \dots\dots\dots(1)$ $\therefore 4 = 2xy \dots\dots\dots(2)$ $(1)^2 + (2)^2$ $x^4 + 2x^2y^2 + y^4 = 25$ $(x^2 + y^2)^2 = 25$ $x^2 + y^2 = 5 \dots\dots\dots(3)$ $(1) + (3) \quad 2x^2 = 8$ $x = \pm 2$ $y = \pm 1$ $\sqrt{A} = \pm(2+i)$	3	1 for squaring and equating real and imaginary 1 for eliminating y (or x) 1 for final solution	
iii)	$A - \bar{B} = 3+4i - (2+2i)$ $= 1+2i$	1	1 for answer	
(b) i)	$1+\sqrt{3}i = 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ $\cos\theta = \frac{1}{2} \quad \text{and} \quad \sin\theta = \frac{\sqrt{3}}{2}$ $\therefore \theta = \frac{\pi}{3}$ $\therefore 1+\sqrt{3}i = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$	2	1 value of θ 1 for result	
ii)	$(1+\sqrt{3}i)^6 = 2^6 \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^6$ $= 64(\cos 2\pi + i\sin 2\pi) \text{ by De Moivre's Theorem}$ $= 64(1+0i)$ $= 64$ Which is totally real.	2	1 Use of De Moivre Thm 1 for answer	

Question 2		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(c) i)	$OA = 2iOC$ $= 2i(p + iq)$ $= -2q + 2pi$	1	1 for answer		
ii)	$OB = OC + OA$ $= (p + iq) + (-2q + 2pi)$ $= (p - 2q) + (2p + q)i$	1	1 for answer		
iii)	$BC = -OA$ $= 2q - 2pi$	1	1 for answer		
iv)	$AC = AB + BC$ $= OC - OA$ $= (p + iq) - (-2q + 2pi)$ $= (p + 2q) + (q - 2p)i$	2	1 for sum of vectors 1 for answer		

Question 3		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(a) i)	Asymptote as $x \rightarrow \infty$ $y = f(x)$ $y = 1$ Minimum $(3, k)$	2	1 for basic shape 1 for asymptote		
ii)	Minimum $(1, 1/3)$ $y = \frac{1}{f(x)}$	2	1 for basic shape 1 for discontinuity		
iii)	$y = -(f(x))^2$ $(1, -9)$	2	1 for shape 1 for below axis		
(b)	$x^2 + 2xy - y^2 = 17$ $2x + 2y + 2x \frac{dy}{dx} - 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx}(2x - 2y) = -(2x + 2y)$ $\frac{dy}{dx} = \frac{-(x + y)}{(x - y)}$ $= \frac{x + y}{y - x}$ At $(3, 2)$ $\text{Gradient of tangent} = \frac{3 + 2}{2 - 3} = -5$	2	1 for implicit differentiation 1 for derivative		

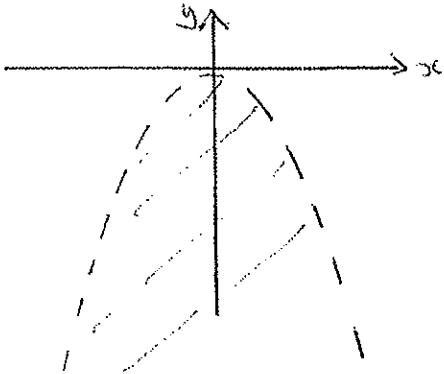
Question 3		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(c) i)	For $x^3 - 3x^2 - 2x + 4 = 0$ $x = \alpha, \beta$ and γ Let $X = x^2$ $\sqrt{X} = x$ $X\sqrt{X} - 3X - 2\sqrt{X} + 4 = 0$ $\sqrt{X}(X - 2) = 3X - 4$ Squaring $X(X^2 - 4X + 4) = 9X^2 - 24X + 16$ $X^3 - 4X^2 + 4X = 9X^2 - 24X + 16$ $\therefore$ Required polynomial is $x^3 - 13x^2 + 28x - 16 = 0$	2	Any method okay. 1 mark for partial solution or complete solution with simple error.		
ii)	As above has roots α^2, β^2 and γ^2 $\alpha^2 + \beta^2 + \gamma^2 = \frac{-b}{a} = 13$	1	1 for answer		
iii)	As α, β and γ are roots of $x^3 - 3x^2 - 2x + 4 = 0$ Then $\alpha^3 - 3\alpha^2 - 2\alpha + 4 = 0$ $\beta^3 - 3\beta^2 - 2\beta + 4 = 0$ $\gamma^3 - 3\gamma^2 - 2\gamma + 4 = 0$ Adding $(\alpha^3 + \beta^3 + \gamma^3) - 3(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha + \beta + \gamma) + 12 = 0$ $(\alpha^3 + \beta^3 + \gamma^3) - 3(13) - 2(3) + 12 = 0$ $(\alpha^3 + \beta^3 + \gamma^3) = 33$	2	Any method okay. 1 mark for partial solution or complete solution with simple error.		
(d)	$P(x) = x^3 + 3x^2 - 24x + k$ $P'(x) = 3x^2 + 6x - 24$ $= 3(x - 2)(x + 4)$ If $P'(x) = 0$ $x = 2, x = -4$ If $x = 2$ is a double zero, $P(2) = (2)^3 + 3(2)^2 - 24(2) + k = 0$ $k = 28$ $P(-4) = (-4)^3 + 3(-4)^2 - 24(-4) + k = 0$ $k = -80$	2	1 possible zeros 1 values of k		

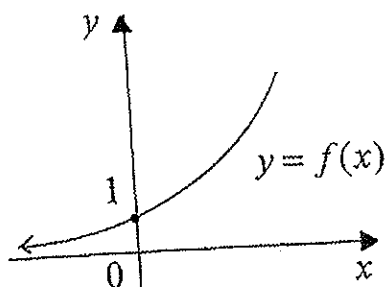
Question 4		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(a) i)	$I_n = \int (\ln x)^n dx = x(\ln x)^n - \int x \cdot n(\ln x)^{n-1} \cdot \frac{1}{x} dx$ $= x(\ln x)^n - n \int (\ln x)^{n-1} dx$ $= x(\ln x)^n - nI_{n-1}$	3	1 for use of Int by parts 1 for simplifying 1 for result in terms of I_n		
ii)	Consider $\int (\ln x)^3 dx = I_3$ $\therefore I_3 = x(\ln x)^3 - 3I_2$ Now $I_2 = x(\ln x)^2 - 2I_1$ and $I_1 = x(\ln x) - I_0$ $= x(\ln x) - x$ $\therefore I_3 = x(\ln x)^3 - 3(x(\ln x)^2 - 2(x(\ln x) - x))$ $= x(\ln x)^3 - 3x(\ln x)^2 + 6x(\ln x) - 6x$ $\therefore \int_1^e (\ln x)^3 dx = (e^4 \cdot 64 - 3e^4 \cdot 16 + 6e^4 \cdot 4 - 6e^4) - (-6)$ $= 34e^4 + 6$	4	1 for I_3 1 for I_2 1 full expression including I_1 1 sub and evaluate		
(b)	$y = 6x - x^2 - 8$ $y = 0 \quad x = 2, 4$ For shells about Y axis $V = 2\pi \int_a^b xy \, dx$ About $x = 1$ $V = 2\pi \int_a^b (x-1)y \, dx$ $= 2\pi \int_2^4 (x-1)(6x - x^2 - 8) \, dx$ $= 2\pi \int_2^4 (7x^2 - x^3 - 14x + 8) \, dx$ $= 2\pi \left[\frac{7x^3}{3} - \frac{x^4}{4} - 7x^2 + 8x \right]_2^4$ $= 2\pi \left[\left(\frac{448}{3} - 64 - 112 + 32 \right) - \left(\frac{56}{3} - 4 - 28 + 16 \right) \right]$ $= \frac{16\pi}{3} \text{ units}^3$	4	4 marks for full solution 3 marks if simple error made 2 marks if major error or 2 simple errors 1 mark if start made using correct formula or from scratch with correct method.		

Question 4		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(c)	 $V_{\text{slice}} \approx Ah \delta y$ $\approx \frac{1}{2} 2x \cdot z \cdot \delta y$ $\approx xz \delta y$	4	1 dimensions of slice		
	$V_{\text{solid}} = \int_{-10}^{10} xz \, dy$ $\frac{x^2}{15^2} + \frac{y^2}{10^2} = 1$ $\therefore x = \sqrt{15^2 \left(1 - \frac{y^2}{10^2}\right)}$ and $\frac{z^2}{15^2} + \frac{y^2}{10^2} = 1$ $\therefore z = \sqrt{15^2 \left(1 - \frac{y^2}{10^2}\right)}$ $\therefore V_{\text{solid}} = 450 \int_0^{10} \left(1 - \frac{y^2}{10^2}\right) dy$ $= 450 \left[y - \frac{y^3}{300} \right]_0^{10}$ $= 3000 \text{ cms}$		1 $x = f(y)$ 1 $z = f(y)$ 1 correct integrand and limits 1 correct exact volume		

Question 5		Trial HSC Examination- Mathematics Extension 2	2008	
Part	Solution	Marks	Comment	
(a) i)	Solving simultaneously $y = mx + a \quad \dots\dots\dots(1)$ $b^2x^2 + a^2y^2 = a^2b^2 \quad \dots\dots(2)$ sub (1) into (2) $b^2x^2 + a^2(mx + a)^2 = a^2b^2$ $b^2x^2 + a^2m^2x^2 + 2a^3mx + a^4 = a^2b^2$ $(b^2 + a^2m^2)x^2 + 2a^3mx + a^4 - a^2b^2 = 0$ If $x = x_1$ and x_2 are the roots then $x_1 + x_2 = \frac{-2a^3m}{b^2 + a^2m^2}$ $\therefore x_2 = \frac{-2a^3m}{b^2 + a^2m^2} - x_1$	3	1 for sub into equation 1 for simplify 1 for expression for x_2	
ii)	For a tangent $x_1 = x_2 = x$ $x_1 + x_2 = 2x = \frac{-2a^3m}{b^2 + a^2m^2}$ $\therefore x = \frac{-a^3m}{b^2 + a^2m^2}$	1	1 for answer	
(b) i)	$x = 2at$ and $y = at^2$ Grad of tangent $= \frac{dy}{dx} = t$ Grad of normal $= -\frac{1}{t}$ At $t = p$ $m = -\frac{1}{p}$ [point $(2ap, ap^2)$] Equation of normal $y - ap^2 = -\frac{1}{p}(x - 2ap)$ $py - ap^3 = -x + 2ap$ $x + py - ap^3 - 2ap = 0$	2	1 for gradient of normal 1 for equation of normal	
ii)	Normal passes through (x_1, y_1) then $x_1 + py_1 - ap^3 - 2ap = 0$ To find intersection with the parabola, this equation must be solved for p. As the equation is a cubic in p, there can be from 1 to 3 values for p. $\therefore$ Up to three normals can be drawn	2	2 for any reasonable explanation	

Question 5		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
(c) i)	$z = \cos \theta + i \sin \theta = c + is$ $z^4 = (c + is)^4$ $= c^4 + 4c^3(is) + 6c^2(-s^2) + 4c(-is^3) + s^4$ $= c^4 - 6c^2s^2 + s^4 + i(4c^3s - 4cs^3)$ By De Moivres Thm $z^4 = \cos 4\theta + i \sin 4\theta$ Equating real parts $\cos 4\theta = c^4 - 6c^2(1 - c^2) + (1 - c^2)^2$ $= c^4 - 6c^2 + 6c^4 + 1 - 2c^2 + c^4$ $= 8c^4 - 8c^2 + 1$ $= 8\cos^4 \theta - 8\cos^2 \theta + 1$	3	1 for expanding 1 for De Moivre 1 for solution		
(ii)	$\left(z + \frac{1}{z}\right)^4 = (2\cos \theta)^4$ $= 16\cos^4 \theta$ and $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3 \frac{1}{z} + 6z^2 \frac{1}{z^2} + 4z \frac{1}{z^3} + \frac{1}{z^4}$ $= z^4 + \frac{1}{z^4} + 4\left(z^2 + \frac{1}{z^2}\right) + 6$ $16\cos^4 \theta = 2\cos 4\theta + 8\cos 2\theta + 6$ $\cos^4 \theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$	2	1 for expansion 1 for expression		
iii)	$\int_0^{\frac{\pi}{2}} \cos^4 \theta \, d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}\right) d\theta$ $= \left[\frac{1}{32}\sin 4\theta + \frac{1}{4}\sin 2\theta + \frac{3}{8}\theta\right]_0^{\frac{\pi}{2}}$ $= \left[\left(\frac{3}{8} \cdot \frac{\pi}{2}\right) - (0)\right]$ $= \frac{3\pi}{16}$	2	1 for integral 1 for evaluating		

Question 6		Trial HSC Examination- Mathematics Extension 2	2008	
Part	Solution	Marks	Comment	
(a)	$[\operatorname{Re}(z)]^2 + \operatorname{Im} z < 0$ $x^2 + y < 0$ $y < -x^2$ 	2	1 for correct Cartesian equation 1 mark for correct region	
b) i)	i. Consider the function $g(x) = e^{-x} + x - 1$. $g(0) = 0$ and $g'(x) = -e^{-x} + 1 \Rightarrow g'(0) = 0$ Also $g''(x) = e^{-x} > 0$ for all x $\therefore g(x)$ has a minimum value of 0 when $x = 0$. $\therefore e^{-x} + x - 1 \geq 0$ for all x, with equality only if $x = 0$. For $x \neq 0$, $f'(x) = \frac{d}{dx} \left(\frac{e^x - 1}{x} \right)$ $= \frac{e^x \cdot x - (e^x - 1) \cdot 1}{x^2}$ $= \frac{1 + xe^x - e^x}{x^2}$ $= \frac{e^x}{x^2} (e^{-x} + x - 1)$ > 0 Hence $f(x)$ is an increasing function for $x \neq 0$.	3	1 shows by differentiation that $e^{-x} + x + 1$ has min value of 0 when $x=0$ 1 finds $f'(x)$ 1 rearranges $f'(x)$ as a product of $e^{-x} + x + 1$ and deduces $f'(x) > 0$	
ii)	ii. Let $h(x) = e^x$. Then $h'(x) = e^x$. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x - e^0}{x - 0}$ $= h'(0)$ $= 1$ $= f(0)$ $\therefore f(x)$ is continuous at $x = 0$.	2	1 Expresses the limiting value of $f(x)$ as $x \rightarrow 0$ as the derivative of e^x at $x=0$ 1 Evaluates this derivative to show limiting value is 1	

Question 6		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
iii)	iii. 	1	Correct shape curve with y intercept of 1 and asymptote		
(c) i)	i. $\frac{dy}{dx} = e^x = 1$ at $x = 0$ Hence tangent AC has equation $y = x + 1$. Let $D(k, 0)$ $\therefore C(k, k+1)$. Also $B(k, e^k)$. Then $\text{Area AODC} < \int_0^k e^x dx < \text{Area AODB}$ $\frac{1}{2}k(k+2) < [e^x]_0^k < \frac{1}{2}k(1+e^k)$ $\therefore \frac{1}{2}k(k+2) < e^k - 1 < \frac{1}{2}k(1+e^k)$ For $k=1$, $1.5 < e - 1 < 0.5 + \frac{1}{2}e$ Hence $2.5 < e$ and $\frac{1}{2}e < 1.5$ $\therefore 2.5 < e < 3$	4	1 for equation of tangent AC and coords of C 1 for lower bound for $e^k - 1$ using area AODC 1 for upper bound using area AODB 1 for using $k=1$ and rearranging		
ii)	ii. Area of $\triangle ABC$ is bisected if $(e^k - 1) - \frac{1}{2}k(k+2) = \frac{1}{2}k(1+e^k) - (e^k - 1)$ $(4-k)e^k - k^2 - 3k - 4 = 0$ Let $f(k) = (4-k)e^k - k^2 - 3k - 4$ Then $f(2) \approx 0.78 > 0$, $f(3) \approx -1.9 < 0$ and $f(k)$ is continuous. Hence $f(k) = 0$, and the area is bisected, for some k such that $2 < k < 3$ $f(k) = (4-k)e^k - k^2 - 3k - 4$ $f'(k) = \{-e^k + (4-k)e^k\} - 2k - 3$ $= (3-k)e^k - 2k - 3$ Taking $k_0 = 2.7$, $k_1 = 2.7 - \frac{f(2.7)}{f'(2.7)}$ $\therefore k_1 \approx 2.7 - \frac{-0.04635}{3.9361}$ ≈ 2.688 Hence second approximation is 2.7 (to one decimal place).	3	1 for equation if triangle area bisected 1 for $f(k)=0$ and establishes existence of root k $2 < k < 3$ 1 for using Newton's method for 2 nd approximation		

Question 7		Trial HSC Examination- Mathematics Extension 2	2008	
Part	Solution	Marks	Comment	
(a) i)	$z^5 = 1$ By De Moivres Thm $\cos 5\theta + i \sin 5\theta = 1$ Equating real and imaginary $\cos 5\theta = 1 \quad \sin 5\theta = 0$ $\therefore 5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$ $\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$ $z_1 = cis 0 = 1$ $z_2 = cis \frac{2\pi}{5}$ $z_3 = cis \frac{4\pi}{5}$ $z_4 = cis \frac{6\pi}{5} = cis \frac{-4\pi}{5} = \bar{z}_3$ $z_5 = cis \frac{8\pi}{5} = cis \frac{-2\pi}{5} = \bar{z}_2$	3	1 values of θ 2 for values of $z_1 - z_5$	
ii)	$z^5 - 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$ $= (z - z_1)(z^2 - (z_2 + z_5)z + z_2z_5)(z^2 - (z_3 + z_4)z + z_3z_4) =$ $= (z - z_1)\left(z^2 - 2\cos\frac{2\pi}{5}z + 1\right)\left(z^2 - 2\cos\frac{4\pi}{5}z + 1\right)$	2	1 factors 1 in quadratics	
iii)	$z^5 - 1 = 0$ $(z - 1)(z^4 + z^3 + z^2 + z + 1) = 0$ Roots are z_2, z_3, z_4, z_5 from above.	1		
iv)	Sum of roots of $z^5 - 1 = 0$ is zero. $\therefore z_1 + z_2 + z_3 + z_4 + z_5 = 0$ $z_1 + z_2 + \bar{z}_2 + z_3 + \bar{z}_3 = 0$ $1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$ $2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = -1$ $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = \frac{-1}{2}$	2	1 for conjugates 1 for answer	

(b)		
i)	$\angle RMA = \angle ABN$ (exterior angle of cyclic quad. $ABNM$ is equal to interior opposite angle) Similarly $\angle ABN = \angle AQP$ in cyclic quadrilateral $ABPQ$. Hence quadrilateral $QAMR$ is cyclic. (exterior angle AQP is equal to interior opposite angle RMA)	3 2 marks for using properties of cyclic quads to deduce equal angles 1 for applying test to $QAMR$
	ii. Produce TM to S. Then $\angle TMR = \angle SMN$ (vertically opposite angles are equal) $\angle SMN = \angle MAN$ (angle between tangent and chord drawn to point of contact is equal to angle subtended by that chord in the alternate segment) $\angle MAN = \angle PAQ$ (vertically opposite angles are equal) $\angle PAQ = \angle TRM$ (exterior angle of cyclic quad. $QAMR$ is equal to interior opposite angle) Hence in $\triangle TMR$, $\angle TMR = \angle TRM$ and hence $TM = TR$ (sides opposite equal angles are equal)	4 1 alt seg theorem 1 vert opp angles 1 equality of angles 1 deduces TMR has equal angles so equal sides

Question 8		Trial HSC Examination- Mathematics Extension 2	2008	
Part	Solution	Marks	Comment	
(a)	$\sin(x+y) - \sin(x-y) = \sin x \cos y + \cos x \sin y - (\sin x \cos y - \cos x \sin y)$ $= 2 \cos x \sin y$	1	1 for answer	
a) ii)	If $n = 1$ LHS = $\cos x$ $\text{RHS} = \frac{\sin\left(\frac{3x}{2}\right) - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$ Using i) above $\text{RHS} = \frac{2 \cos x \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$ $= \cos x = \text{LHS}$ $\therefore$ true for $n = 1$ Assume true for $n = k$ $\text{i.e. } \cos x + \cos 2x + \cos 3x + \dots + \cos kx = \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$ When $n = k + 1$ $\cos x + \cos 2x + \cos 3x + \dots + \cos kx + \cos(k+1)x = \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} + \cos(k+1)x$ $= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \cos(k+1)x \cdot 2 \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$ $= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \sin\left((k+1)x + \frac{x}{2}\right) - \sin\left((k+1)x - \frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)} \quad \text{using i) above}$ $= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right) + \sin\left(k + \frac{3}{2}\right)x - \sin\left(k + \frac{1}{2}\right)x}{2 \sin\left(\frac{x}{2}\right)}$ $= \frac{\sin\left((k+1) + \frac{1}{2}\right)x - \sin\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}$ $\therefore$ True for $n = k + 1$ $\therefore$ Since true for $n = 1$, by induction is true for all positive integral values of $k \geq 1$	4	1 for $n=1$ case 1 for using i) 1 for simplifying 1 for stating $k+1$ case	

Question 8		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution	Marks	Comment		
a) ii)	$\cos 2x + \cos 4x + \cos 6x + \dots + \cos 16x = \cos(2x) + \cos 2(2x) + \cos 3(2x) + \dots + \cos 8(2x)$ $= \frac{\sin\left(8 + \frac{1}{2}\right)2x - \sin\left(\frac{2x}{2}\right)}{2\sin\left(\frac{2x}{2}\right)}$ $= \frac{\sin 17x - \sin x}{2\sin x}$ $= \frac{\sin(9+8)x - \sin(9-8)x}{2\sin x}$ $= \frac{2\cos 9x \sin 8x}{2\sin x} \quad \text{Using i) above}$ $= \frac{2\cos 9x \cdot 2\sin 4x \cos 4x}{2\sin x} \quad \text{Using double angle on } \sin 8x$ $= \frac{4\cos 9x \cdot 2\sin 2x \cos 2x \cos 4x}{2\sin x} \quad \text{Using double angle on } \sin 4x$ $= \frac{8\cos 9x \cdot 2\sin x \cos x \cos 2x \cos 4x}{2\sin x} \quad \text{Using double angle on } \sin 2x$ $= \frac{8\cos 9x \cdot \cancel{2\sin x} \cos x \cos 2x \cos 4x}{\cancel{2\sin x}}$ $= 8\cos 9x \cos 4x \cos 2x \cos x$	4	1 for sub into expression 1 for breaking up 17x 2 for completing simplification		
(b)	Let roots be $\alpha - d$, α and $\alpha + d$ $\therefore$ Sum of the roots $= (\alpha - d) + \alpha + (\alpha + d) = -a$ $\therefore 3\alpha = -a$ $\alpha = \frac{-a}{3}$ $\therefore$ Sum of the roots 2 at a time $= (\alpha - d)\alpha + (\alpha - d)(\alpha + d) + (\alpha + d)\alpha = b$ $\alpha^2 - \alpha d + \alpha^2 - d^2 + \alpha^2 + \alpha d = b$ $3\alpha^2 - d^2 = b$ $d^2 = 3\alpha^2 - b$ $d^2 = 3\left(\frac{-a}{3}\right)^2 - b = \frac{a^2}{3} - b$ $\therefore$ Product of the roots $= \alpha(\alpha - d)(\alpha + d) = c$ $\alpha^3 - \alpha d^2 = c$ $\left(\frac{-a}{3}\right)^3 - \left(\frac{-a}{3}\right)\left(\frac{a^2}{3} - b\right) = c$ $\frac{-a^3}{27} + \frac{a^3}{9} - \frac{ab}{3} = c$ $\frac{2a^3}{27} - \frac{ab}{3} = c$ $\therefore 2a^3 - 9ab - 27c = 0$	4	1 each for expressions for sums & products = 3 marks 1 for substitution and simplifying		

Question 8		Trial HSC Examination- Mathematics Extension 2		2008	
Part	Solution			Marks	Comment
(c)	$\frac{dy}{dx} = 2y$ $\therefore \frac{dx}{dy} = \frac{1}{2y}$ $\therefore x = \frac{1}{2} \ln y + c$ When $x = 1, y = 1$ $\therefore c = 1$ $\therefore x = \frac{1}{2} \ln y + 1$ $\frac{1}{2} \ln y = x - 1$ $\ln y = 2x - 2$ $y = e^{2x-2}$			2	1 for expression for x 1 for result

